

Exercise Sheet 1

Introduction

Exercise 1.1 — Safety vs Security

Difference between Safety and Security

Safety refers to protection against accidental failures that may cause harm to people, environment, or property.

Security refers to protection against intentional attacks or malicious actions.

Example of Safety Concern

For an autonomous vehicle, a safety concern is **failure to detect a pedestrian**, causing the vehicle not to brake and potentially resulting in an accident.

Example of Security Concern

A security concern is an **adversarial attack** where an attacker modifies road signs or camera inputs to fool the perception model.

Exercise 1.2 — EU AI Act (*Optional*)

1. Risk Categories

Category	Description
Unacceptable Risk	AI systems prohibited because they create unacceptable harm
High Risk	AI systems affecting safety or fundamental rights
Limited Risk	Systems requiring transparency obligations
Minimal Risk	Low-risk applications

2a. Autonomous Vehicle Perception System

Category:

High Risk

Reason:

The perception system directly influences driving decisions and failures may endanger human life.

2b. ChatBot

Category:

Limited Risk

Reason:

ChatBots generally require transparency obligations but usually do not directly affect safety.

3. Obligations for High-Risk Systems

Providers must ensure:

- Risk management
 - Data governance
 - Documentation
 - Transparency
 - Human oversight
 - Accuracy and robustness evaluation
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Exercise 1.3 — Build the Right Model vs Build the Model Right

Build the Right Model (Correctness)

Failure example:

The pedestrian detector does not recognize children crossing the road because the training data lacked such examples.

This means the model solves the wrong problem.

Build the Model Right (Robustness)

Failure example:

The model performs well during sunny weather but fails during fog or night driving.

This indicates insufficient robustness.

Exercise 1.4 — First Impressions

1. Three situations where model predictions cannot be trusted

Night driving

Training data only contains daytime images. Performance at night is unknown.

Fog or rain

Weather changes visibility and may reduce detection performance.

Camera obstruction

Mud, glare or dirt on the camera may produce incorrect predictions.

2. Most safety-critical label

Pedestrian present

Reason:

Missing a pedestrian may directly lead to injury or loss of life.

Vehicle and traffic light failures are also important, but pedestrian safety is most critical.

3. Additional information required before deployment

Before deployment:

- Performance under fog/night conditions
 - Recall values
 - ODD coverage
 - Failure analysis
 - Robustness evaluation
 - Distribution shift analysis
 - Human intervention procedures
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Exercise 1.5 — Course Topics and Safety Case (*Optional*)

Topic	Contribution
Testing & Validation	Measures model performance
Explainability	Explains model decisions
Uncertainty Estimation	Measures prediction confidence
Anomaly Detection	Detects OOD conditions
Adversarial ML	Tests robustness against attacks
Alignment	Ensures system goals match safety objectives